

GHL Clone Platform

Phase 1 — Product Specification

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7	4	3	17
Modules	Months	Developers	Sprints Est.

1.1 Dashboard & Analytics	1.2 Contact Management	1.3 Lead Management	1.4 Marketing Campaigns	1.5 Automation Engine	1.6 Conversation Inbox	1.7 Super Admin & RBAC
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Pre-Phase 1: Foundation Sprint (Weeks 1–2)

Before feature development begins, the following foundational work must be completed. All three developers work on this in parallel. Nothing else starts until this is solid.

All developers must be involved in the foundation sprint. No feature work begins until auth, RBAC, and multi-tenancy are complete and tested.

Authentication

- JWT access tokens + refresh token rotation
- Bcrypt password hashing
- Token blacklist in Redis (for logout and revocation)
- Session management middleware on all protected routes

RBAC — Role-Based Access Control

- Role hierarchy: Super Admin → Agency Admin → Sub-account Admin → Team Member
- Permission table per role per module (view, create, edit, delete)
- RBAC middleware applied globally on all API routes
- Custom role scaffolding (fully expanded in Phase 3)

Multi-tenancy

- tenant_id on every database table — enforced at ORM (Prisma) level
- Prisma middleware to auto-inject tenant context on every query
- Subdomain routing — each sub-account resolves to a subdomain
- Automated tenant isolation tests from Day 1, run on every CI build

Infrastructure

- Turborepo monorepo: apps/web (Next.js), apps/api (Fastify), packages/db (Prisma), packages/types
- CI/CD pipeline via GitHub Actions — auto-deploy on merge to main
- Environments: dev, staging, production
- PostgreSQL (Neon) + Redis (Upstash) setup and initial migrations
- Cloudflare R2 bucket setup and SDK integration
- Error monitoring (Sentry) and structured logging (Pino)

Phase 1 — Overview

Phase 1 delivers the core CRM, multi-channel marketing campaigns, lead management, workflow automation, chatbot builder, and unified conversation inbox. It is the MVP that enables the first paying clients.

#	Module	Weeks	Primary Owner	Dependencies
1.1	Dashboard & Analytics	Wk 3–6	Frontend Dev	All modules (shell first)
1.2	Contact Management	Wk 3–5	Backend + FS	Foundation
1.3	Lead Management	Wk 5–7	Backend Dev	1.2 Contacts
1.4	Marketing Campaigns	Wk 5–11	Full-stack Dev	1.2 Contacts
1.5	Automation Engine	Wk 10–15	Full-stack Dev	1.2, 1.3, 1.4
1.6	Conversation Inbox	Wk 11–14	Backend + FE Dev	1.2, 1.4

WhatsApp Meta Business API application must be submitted on Day 1. Meta approval takes 24–72 hours minimum and is a dependency for modules 1.4 and 1.6. Also begin DLT registration for SMS (India) on Day 1.

1.1

Dashboard & Analytics

Central hub for real-time business metrics across all modules

The dashboard is a consumer module — it reads from all other modules. Build the shell and layout in Weeks 3–4 with placeholder widgets, which are backfilled with real data as each sprint completes.

1.1.1 KPI Widgets

- Total Contacts — count with week-on-week change indicator
- Active Leads — total open deals and total pipeline value
- Campaigns Sent — count this month with delivery rate %
- Messages Delivered — breakdown by channel: Email / SMS / WhatsApp
- Automation Runs — total active workflows and executions this month
- Conversations Open — unread count from inbox

1.1.2 Charts & Analytics

- Contact Growth Chart — new contacts over time (line chart, 30/60/90 day)
- Campaign Performance Summary — open rate, click rate, delivery rate per channel
- Lead Funnel Chart — contacts → leads → pipeline stages → won (funnel chart)
- Message Volume Chart — daily send volume by channel (bar chart)

1.1.3 Activity & Navigation

- Recent Activity Feed — real-time log: new contacts, campaign sends, automation triggers, new leads
- Quick Actions Panel — Create Contact, Launch Campaign, New Lead, New Automation
- Date Range Selector — Today / Last 7 days / Last 30 days / Custom range
- Multi-account View (Agency) — aggregate stats across all sub-accounts in one view

Dashboard widgets are built as independent components from day one so future modules (Phase 2+) can plug in new data without rework.

1.2

Contact Management

Full lifecycle contact management with custom fields, segments, and activity tracking

The contact is the core entity of the entire platform. Every module — campaigns, leads, automations, inbox — links back to a contact. The schema and API must be finalised before other modules begin.

1.2.1 Contact Records

- Create, read, update, delete contacts
- Standard fields: First Name, Last Name, Email, Phone, Company, Address, Date of Birth, Source
- Custom fields — types: Text, Number, Date, Dropdown, Multi-select, Checkbox, URL
- Custom fields stored as JSONB on PostgreSQL — flexible per tenant, no schema migrations needed
- Contact owner — assign to a team member
- Source tracking — manual, import, API, campaign, form submission
- Duplicate detection — match by email or phone on create and import
- Contact profile page — all fields, activity timeline, linked deals, sent campaigns

1.2.2 Segmentation & Lists

- Tags — apply multiple tags per contact, filter by tag, bulk tag/untag
- Static Lists — manually curated groups, add/remove contacts
- Smart Lists — dynamic filter-based segments that auto-update as contacts match criteria
- Bulk actions — tag, assign, add to campaign, export, delete
 - Example Smart List: tag = 'hot-lead' AND created_at > 30 days ago AND city = 'Mumbai'

1.2.3 Import & Export

- CSV import with column-to-field mapping UI
- Import preview — show first 5 rows before confirming
- Duplicate handling on import — skip / merge / create duplicate options
- Background import job (BullMQ) for large files with progress indicator
- CSV export — export any filtered view or smart list with field selector

1.2.4 Activity Timeline

- Chronological log of all contact interactions
- Event types: Email Sent/Opened/Clicked, SMS Sent/Delivered, WhatsApp Sent/Read, Deal Created, Stage Changed, Tag Added, Automation Triggered, Note Added, Form Submitted
- Manual notes — agents can add timestamped notes with rich text
- Timeline filter — filter by event type
- Linked to inbox — messages from inbox appear in timeline automatically

1.3**Lead Management**

Pipeline-based lead and deal tracking from capture through close

Leads in Phase 1 use a single default pipeline per sub-account. Multiple pipelines and advanced forecasting are delivered in Phase 2. Focus is a clean, fast Kanban board.

1.3.1 Pipeline Board

- Kanban board — drag-and-drop cards between stages
- Default stages: New Lead → Contacted → Qualified → Proposal Sent → Won / Lost
- Custom stage names and ordering — editable per sub-account
- Deal card preview — contact name, deal value, assigned user, days in current stage
- Quick edit on card — update stage, value, or assignee without opening full deal
- List view toggle — switch between Kanban and table view
- Filters — by assigned user, stage, source, date range

1.3.2 Deal Fields

- Deal name, linked contact, deal value (INR / USD configurable)
- Expected close date and priority (Low / Medium / High)
- Assigned team member and lead source
- Won / Lost marking with reason dropdown
- Custom fields on deals using same JSONB system as contacts

1.3.3 Lead Capture & Activity

- Manual lead creation from pipeline board or contact record
- Automation action — 'Create Deal' node in workflow builder creates a lead automatically
- API endpoint — external tools can POST to create leads programmatically
- Full activity log per deal — stage changes, notes, emails sent
- Internal notes on deal (separate from contact timeline)

1.4

Marketing Campaigns

Multi-channel campaign builder — Email, SMS, and WhatsApp

All three channels share a common campaign model and targeting system. Send engines are built behind a provider abstraction interface — swapping providers is a config change, not a code change.

1.4.1 Common Campaign System

- Campaign model: name, channel, status, audience, schedule, created_by, tenant_id
- Campaign statuses: Draft → Scheduled → Sending → Sent / Failed
- Audience targeting — contact list, tag, smart list, or manual contact selection
- Schedule options — send immediately or schedule for a specific date and time
- Campaign duplication — clone any campaign as a starting point

1.4.2 Email Campaigns

Template Builder

- Drag-and-drop email builder via Unlayer (embedded SDK — do not build from scratch)
- Saved template library at agency and sub-account level
- Merge tags: {{first_name}}, {{last_name}}, {{company}}, {{phone}}, custom field tags
- Plain-text version auto-generated alongside HTML
- Mobile preview toggle in the editor

Send & Tracking

- From name, from email, reply-to, subject line and preview text
- Send via Resend (primary) / AWS SES (fallback)
- Open tracking (pixel), click tracking (link wrapping)
- Delivery webhooks: delivered, bounced, spam, unsubscribed
- Per-campaign report: sent, delivered %, opened %, clicked %, bounced %, unsubscribed %
- Unsubscribe link auto-injected — global opt-out list per tenant

1.4.3 SMS Campaigns

- SMS composer with real-time character count and segment indicator (160 chars = 1 SMS)
- Merge tags support in SMS body
- Provider: MSG91 (primary, India) / Twilio (fallback / international)
- Sender ID configuration per sub-account
- DLT template ID field on templates (required for Indian SMS compliance)
- Delivery status tracking via webhook: queued, sent, delivered, failed
- Opt-out handling — STOP/UNSUBSCRIBE keyword auto-adds to opt-out list
- SMS template library — save and reuse message templates

1.4.4 WhatsApp Campaigns

Meta Integration

- Meta Cloud API integration — WABA per sub-account
- Phone number registration and verification within the platform
- Message template sync — pull all Meta-approved templates into platform
- Template submission workflow — submit new templates to Meta for approval from within the platform

Campaign Send & Tracking

- Campaign builder using approved templates only (Meta policy requirement)
- Audience targeting with variable preview per contact
- Delivery tracking via Meta webhooks: sent, delivered, read
- Failed message logging with Meta error codes
- Incoming reply capture — logged to contact activity timeline and conversation inbox
- Opt-out handling — STOP reply marks contact as WhatsApp unsubscribed

1.5

Automation Engine

Visual workflow builder and flow-based chatbot builder

The most complex module in Phase 1. Uses React Flow for the drag-and-drop canvas and BullMQ for job execution. A single canvas component is shared between the workflow builder and chatbot builder.

Use React Flow for the canvas — do not build from scratch. The same canvas component serves both the workflow builder and chatbot builder, saving 3–4 weeks of frontend development.

1.5.1 Workflow Builder

Triggers (Phase 1)

- Contact Created, Tag Added, Tag Removed
- Campaign Event — email opened, email clicked, SMS delivered, WhatsApp read
- Deal Stage Changed — fires when a deal moves to a specified stage
- Form Submitted — stub for Phase 3 form builder
- Time Delay — wait X minutes / hours / days before next step
- Scheduled Trigger — run at a specific date/time (one-off or recurring)

Conditions

- If / Else branch — split flow based on a condition
- Condition types: contact field value, tag exists/absent, deal stage, channel engagement
- Multi-condition logic — AND / OR combinations

Actions (Phase 1)

- Send Email / Send SMS / Send WhatsApp (template)
- Add Tag / Remove Tag
- Update Contact Field — set field to specified value
- Move Deal Stage — move contact's linked deal to a specified stage
- Assign Contact to a team member
- Create Deal — create a new deal linked to the contact
- Internal Notification — send in-app notification to a team member
- Webhook — POST contact data to an external URL (outbound integration)

Execution & Monitoring

- BullMQ job queue per action type — independent scaling and failure isolation
- Per-contact execution history — which steps ran, when, and outcome
- Workflow run log — total runs, success rate, failed steps
- Error notifications — notify workflow owner on repeated failures
- Rate limiting — max executions per contact per hour to prevent infinite loops

1.5.2 Chatbot Builder

Node Types

- Send Message — bot sends a text message to the contact
- Ask Question — bot asks a question, waits for any reply
- Show Options — numbered or button options (WhatsApp list messages)
- Capture Input — saves user's reply to a specified contact custom field
- Condition — branch based on captured input value or contact field
- Set Field — update contact field based on conversation data
- Human Handoff — transfers conversation to an agent in the Conversation Inbox
- End Conversation — closes the chatbot session for this contact

Activation, Channels & Sessions

- Keyword trigger — chatbot activates when contact sends a specific keyword (e.g. 'HI', 'START', 'BOOK')
- Channels: WhatsApp (primary), SMS (fallback for simple linear flows)
- Test mode — simulate a full conversation in the builder before publishing
- Active sessions view — see all contacts currently mid-conversation
- Session timeout — auto-end chatbot session after configurable inactivity period
- Conversation log — full transcript per contact session with timestamps

1.6**Unified Conversation Inbox**

All customer conversations across Email, SMS and WhatsApp in one view

The conversation inbox consolidates all inbound and outbound messages across Email, SMS, and WhatsApp into a single threaded interface. Agents never need to switch between channel tools. Every conversation links back to a contact record.

1.6.1 Inbox Layout

- Split-pane layout — conversation list on left, active conversation thread on right
- Contact mini-profile panel on far right — name, tags, key fields, linked deal
- Unread count badge per channel filter and overall inbox
- Real-time updates via WebSocket (Socket.io) — new messages appear without page refresh
- Channel indicator badge per conversation (Email / SMS / WA icon)

1.6.2 Conversation Management

- Conversation statuses: Open, Resolved, Snoozed
- Snooze — hide conversation until a set date/time, then automatically resurface
- Assign conversation to a team member
- Search conversations — by contact name, phone, email, or message content
- Filter — by channel, status, assigned agent, unread only, date range
- Sort — by most recent / oldest / unread first

1.6.3 Compose & Reply

- Channel selector — choose to reply via Email, SMS, or WhatsApp per message
- Rich text for email replies (bold, italic, links, attachments)
- Character counter for SMS, template picker for WhatsApp outbound messages
- Merge tags in replies — insert contact field values dynamically
- File attachments — images and documents for WhatsApp and Email
- Send and mark resolved in one action
- Internal notes — visible to agents only, supports @mention team members
- @mention triggers in-app notification to the mentioned agent

1.6.4 Canned Responses & Notifications

- Canned responses — create and manage saved reply templates per channel
- Search canned responses while composing — type '/' to trigger search
- Agency-level shared + sub-account-level private canned responses
- In-app notification bell — new messages, assignments, @mentions (real-time via WebSocket)
- Email notification for new conversation assignment (configurable per user)
- Browser push notifications (opt-in)

1.7

Super Admin & RBAC

Platform-wide administration, role hierarchy, agency management, and white-label configuration

The Super Admin module is the control centre of the entire platform. It sits above all tenants and manages the agency layer. Every role, permission, agency account, and white-label configuration is controlled from here. This module is foundational — it must be built in the foundation sprint before any other feature work begins.

RBAC and multi-tenancy are built in Weeks 1–2 (Foundation Sprint). The Super Admin UI panel is built in Weeks 3–5 in parallel with Contact development. White-label config UI ships in Week 6 alongside the Agency Portal.

1.7.1 Role Hierarchy

The platform enforces a strict 5-level role hierarchy. Each level can only manage roles below it. Permissions flow downward and cannot be escalated by a lower role.

Role	Scope	Key Capabilities
Super Admin	Platform-wide	Create/suspend agencies, global config, billing oversight, platform announcements, impersonate any user for support
Agency Admin	Agency + all sub-accounts	Create sub-accounts, assign team members, configure white-label branding, view all sub-account data
Sub-account Admin	Single sub-account	Full access within their sub-account — manage contacts, campaigns, automations, inbox, team members
Team Member	Single sub-account	Assigned permissions only — e.g. can manage inbox but not campaigns, based on role config
Custom Role	Single sub-account	Granular permission set defined by Sub-account Admin — mix and match module access

1.7.2 Super Admin Panel

Accessible only to Super Admin users. A separate admin portal (e.g. admin.yourdomain.com) distinct from the agency-facing UI.

- **Agency list** — view all registered agencies, status (Active / Suspended / Trial), plan, creation date
- **Create agency** — provision a new agency account with name, owner email, plan, and custom domain
- **Suspend / reactivate agency** — suspend all logins and API access for a given agency
- **Impersonate** — log in as any agency admin or sub-account user for support debugging (audit logged)
- **Platform announcements** — broadcast in-app banner messages to all agencies or specific ones
- **Global feature flags** — enable/disable modules per agency (e.g. WhatsApp not yet available for a specific agency)
- **Usage stats** — contacts count, campaign sends, active automations per agency

- **Audit log** — every Super Admin action logged with timestamp, actor, and target

1.7.3 Agency Management (Agency Admin View)

Agency Admins manage their own sub-accounts and team members from the Agency Portal.

- **Sub-account list** — view all sub-accounts under the agency, status, last activity
- **Create sub-account** — provision a new sub-account with name, industry, owner assignment
- **Suspend sub-account** — temporarily disable login and access for a sub-account
- **Switch to sub-account** — agency admin can jump into any sub-account as an admin (audit logged)
- **Team members** — invite team members to the agency level or directly to a sub-account
- **Seat management** — view how many team member seats are used across sub-accounts
- **Agency-level settings** — default timezone, date format, notification preferences

1.7.4 RBAC Permissions Engine

Permissions are stored per role per sub-account. The backend enforces them on every API request — client-side role checks are decorative only.

Module	Granular Permissions
Contacts	view_contacts, create_contact, edit_contact, delete_contact, export_contacts, import_contacts
Leads	view_pipeline, manage_deals, delete_deals
Campaigns	view_campaigns, create_campaign, send_campaign, delete_campaign
Automation	view_workflows, create_workflow, activate_workflow, delete_workflow, manage_chatbots
Inbox	view_inbox, reply_conversation, assign_conversation, resolve_conversation
Reports	view_reports, export_reports
Settings	manage_team, manage_roles, manage_integrations, manage_billing

- **Custom role builder** — Sub-account Admins pick individual permissions from the table above to create named roles (e.g. 'Sales Rep', 'Support Agent')
- **Role assignment** — assign one role per team member per sub-account
- **Permission inheritance** — Agency Admin inherits all sub-account permissions automatically, no config needed
- **Backend enforcement** — Fastify permission middleware runs on every protected route, reads role from JWT, checks DB permissions
- **Permission cache** — role → permission map cached in Redis (TTL 5 min) to avoid DB hit on every request

1.7.5 White-Label Configuration (Agency Admin)

Each agency can fully white-label the platform for their clients. Sub-accounts see the agency's brand, not PhantomCore's brand.

Agency Name	Displayed in the browser tab, emails, and login page header
Logo	Light mode + dark mode logo variants — stored in Cloudflare R2, served via CDN
Favicon	16×16 / 32×32 favicon for browser tab
Primary Colour	Hex colour for buttons, active states, links — applied via CSS variable
Custom Domain	e.g. app.youragency.com → routed via Cloudflare CNAME to platform servers
Email Sender Name	e.g. 'Notifications from YourAgency' — used in all system emails
Email Sender Domain	Custom sending domain (e.g. mail.youragency.com) verified via DNS (SPF, DKIM)
Login Page Copy	Custom headline and tagline on the sub-account login page
Support Email	Shown in footer of system emails and error pages — routes to agency's support
Hide Powered By	Option to remove all PhantomCore branding from the UI (Phase 1 optional)

Custom domain routing works via Cloudflare: agency sets a CNAME record pointing to the platform. The API reads the Host header on every request, looks up the agency by domain in Redis/DB, and loads the correct branding config. SSL is handled by Cloudflare automatically.

Sprint Plan — 4 Month Delivery

Development runs across three parallel tracks. Adjust weeks based on team velocity after the first sprint review.

Weeks	Backend Track	Frontend Track	Integration / DevOps
Wk 1–2	Foundation: Auth, RBAC engine, multi-tenancy, DB schema, Super Admin data model	Project setup, design system, component library, Super Admin portal shell	Monorepo, CI/CD, AWS infra setup, Docker + Nginx config
Wk 3–5	Super Admin panel APIs + Agency CRUD; Contact APIs, custom fields, segments	Super Admin UI (agency list, create, suspend); Contact list, detail view	CSV import worker (BullMQ), submit Meta WABA application
Wk 5–7	White-label config API, custom domain resolver; Lead / deal APIs, pipeline stages	Agency Portal white-label settings UI; Kanban board, deal cards, dashboard shell	Cloudflare domain routing, CNAME setup, dashboard widget API contracts
Wk 6–9	Email + SMS campaign engines, provider abstraction, webhooks	Campaign builder UI, Unlayer email editor, SMS composer	Resend + MSG91 integration, delivery webhooks
Wk 8–11	WhatsApp campaign engine, Meta API, template sync	WhatsApp builder UI, template picker, delivery report	Meta webhook verification, R2 storage
Wk 10–14	Automation engine, BullMQ workflow execution, chatbot session engine	React Flow canvas, workflow + chatbot node panels	Automation test harness, rate limit config
Wk 11–14	Conversation inbox API, WebSocket server (Socket.io), canned responses	Inbox layout, compose UI, real-time updates	Socket.io + Redis adapter, push notifications
Wk 15–17	Bug fixes, performance, query optimisation, API hardening, RBAC audit	Dashboard backfill, mobile responsive, UX polish	Load testing, tenant isolation audit, staging deploy

Risks & Mitigations

Risk	Impact	Mitigation
WhatsApp Meta API approval delays	High	Submit WABA application Day 1. Use Meta sandbox for development from Week 3. Never start WhatsApp work while waiting for full approval.
Automation engine scope creep	High	Limit Phase 1 strictly to listed triggers and actions. Any additions go on Phase 2 backlog. No exceptions during sprints.
Tenant data bleed between sub-accounts	Critical	Write automated tenant isolation tests from Week 2. Run on every CI build. Test with 3 simultaneous tenants in staging environment.
Unlayer email builder integration issues	Medium	Spike integration in Week 3 before campaign sprint begins. Have plain-text editor as emergency fallback.
React Flow canvas performance with large flows	Medium	Limit visible nodes, implement virtualization if flow exceeds 50 nodes. Test with realistic data early.
DLT registration delays for SMS (India)	Medium	Start DLT registration on Week 1 in parallel with development. Use international Twilio numbers in testing environment.
Real-time inbox message delays	Medium	Use Socket.io with Redis adapter for multi-instance support. Load test WebSocket connections in Week 14.

Out of Scope — Phase 1

The following items are explicitly excluded from Phase 1. Any request to include them during Phase 1 sprints must be escalated to the product owner and placed on the Phase 2 backlog.

- Social media post scheduling and management
- AI content generation (captions, email copy, SMS copy)
- Calendar and appointment booking
- Reputation and review management
- Telephony and call centre integration
- Payments, invoicing, and subscriptions
- Funnel and landing page builder
- White-label agency portal and custom domains
- Multiple pipelines — single default pipeline only in Phase 1
- Advanced custom role creation — roles are pre-defined in Phase 1

Scope changes during Phase 1 sprints will be logged, estimated, and scheduled for Phase 2. No unplanned features will be added mid-sprint under any circumstances.